A* Heuristic Analysis: Scaled Manhattan (Harder Version)

Advanced exam-style question on admissibility, consistency, and dominance

Problem Setup

Goal: Bucharest B at coordinate $(0, 0)$.

Define Scaled Manhattan Heuristic

Define:

$$h_k(n) = k \cdot h_{\text{MAN}}(n), \qquad 0 < k \leq 1,$$

where

$$h_{\text{MAN}}(n) = |x_n| + |y_n|.$$

Given Data

Coordinates

- $F(-160, 78)$
- $R(-168, 95)$
- $P(-70, 69)$
- $S(-200, 155)$
- $B(0, 0)$

Pre-computed Manhattan values

Node	$h_{\text{MAN}}(n)$
F	238
R	263
P	139
S	355
B	0

Edge costs

- $F \leftrightarrow B = 211$
- $R \leftrightarrow P = 97$
- $S \leftrightarrow R = 80$

SLD heuristic value (given)

$$h_{\text{SLD}}(F) = 178.$$

Questions

QH1. Find largest k for admissibility at F

The true optimal cost from F to B is 211 (via the direct edge). Find the largest value of k such that $h_k(F)$ is admissible.

Recall: admissibility requires $h(n) \leq h^*(n)$.

QH2. Find largest k for consistency on each edge

Using the consistency inequality

$$h(n) \leq c(n, n') + h(n'),$$

find the largest value of k such that h_k is consistent on:

1. Edge $R \to P$ (cost 97)
2. Edge $S \to R$ (cost 80)
3. Edge $F \to B$ (cost 211)

For each edge, derive the constraint on k and compute the numerical upper bound.

QH3. Determine k^* that satisfies all constraints

Let k^* be the largest value that satisfies all constraints from QH1 and QH2 simultaneously. Compute k^* (exact fraction and decimal approximation).

QH4. Compare with SLD and discuss dominance

1. Using k^* from QH3, compute $h_{k^*}(F)$.
2. Compare with $h_{\text{SLD}}(F) = 178$. Which is larger at node F?
3. Can you conclude that h_{k^*} dominates h_{SLD} on the entire graph based only on this comparison? Explain carefully.

Complete Worked Solution

Solution to QH1 (Admissibility at F)

Admissibility requires:

$$h_k(F) \leq h^*(F).$$

We know:

$$h^*(F) = 211, \qquad h_k(F) = k \cdot 238.$$

Therefore:

$$k \cdot 238 \leq 211 \quad \Longrightarrow \quad k \leq \frac{211}{238} \approx 0.886554621.$$

Answer QH1: $k \leq \dfrac{211}{238} \approx 0.8866.$

Solution to QH2 (Consistency constraints)

For each edge $n \to n'$:

$$h_k(n) \le c(n, n') + h_k(n')$$
$$k\, h_{\text{MAN}}(n) \le c(n, n') + k\, h_{\text{MAN}}(n')$$
$$k\big(h_{\text{MAN}}(n) - h_{\text{MAN}}(n')\big) \le c(n, n').$$

When $h_{\text{MAN}}(n) > h_{\text{MAN}}(n')$ this implies:

$$k \le \frac{c(n, n')}{h_{\text{MAN}}(n) - h_{\text{MAN}}(n')}.$$

(1) Edge $R \to P$ (cost 97).

$$h_{\text{MAN}}(R) - h_{\text{MAN}}(P) = 263 - 139 = 124,$$

so

$$k \le \frac{97}{124} \approx 0.782258065.$$

Constraint 1: $k \le \dfrac{97}{124} \approx 0.7823$.

(2) Edge $S \to R$ (cost 80).

$$h_{\text{MAN}}(S) - h_{\text{MAN}}(R) = 355 - 263 = 92,$$

so

$$k \le \frac{80}{92} = \frac{20}{23} \approx 0.869565217.$$

Constraint 2: $k \le \dfrac{20}{23} \approx 0.8696$.

(3) Edge $F \to B$ (cost 211).

$$h_{\text{MAN}}(F) - h_{\text{MAN}}(B) = 238 - 0 = 238,$$

so

$$k \le \frac{211}{238} \approx 0.886554621.$$

Constraint 3: $k \le \dfrac{211}{238} \approx 0.8866$.

Solution to QH3 (Combined k^*)

All constraints must hold simultaneously. Summarize the upper bounds:

Source	Upper bound on k
Admissibility at F	$k \le \dfrac{211}{238} \approx 0.8866$
Consistency $R \to P$	$k \le \dfrac{97}{124} \approx 0.7823$
Consistency $S \to R$	$k \le \dfrac{20}{23} \approx 0.8696$
Consistency $F \to B$	$k \le \dfrac{211}{238} \approx 0.8866$

The tightest constraint is from edge $R \to P$, hence:

$$k^* = \frac{97}{124} \approx 0.7823.$$

Answer QH3: $k^* = \dfrac{97}{124} \approx 0.7823$.

Solution to QH4 (Comparison with SLD)

1) Compute $h_{k^*}(F)$.

$$h_{k^*}(F) = k^*\, h_{\mathrm{MAN}}(F) = \left(\frac{97}{124}\right) \cdot 238 = \frac{97 \cdot 238}{124} = \frac{23086}{124} \approx 186.1774.$$

So:

$$h_{k^*}(F) \approx 186.18.$$

2) Compare with $h_{\mathbf{SLD}}(F) = 178$.

$$186.18 > 178,$$

so at node F:

$$h_{k^*}(F) > h_{\mathrm{SLD}}(F).$$

3) Can we conclude dominance? No. Dominance is a *global* property:

- It requires $h_{k^*}(n) \geq h_{\mathrm{SLD}}(n)$ for *all* nodes n.
- Both heuristics must be admissible.

Showing the inequality at only one node (F) is insufficient.

Answer QH4: We cannot conclude that h_{k^*} dominates h_{SLD} based only on the comparison at node F.